

Customer Shopping Behavior Analysis

1. Project Overview

This project focuses on analyzing 4,500 customer records to uncover meaningful insights into customer purchasing behavior, sales trends, product performance, customer segmentation, and subscription patterns. The analysis helps identify key business trends and supports data-driven decision-making.

2. Dataset Summary

Rows: 4,500

Columns: 20

Calculated Column: Customer Segment *(created in Power BI)*

Key Features:

Customer details (Age, Gender, Location, Subscription Status)

Purchase details (Item Purchased, Category, Purchase Amount INR, Season, Size, Color, Unit sold, Unit price)

Shopping behavior (Discount Applied, Previous Purchases, Frequency of Purchases, Payment Method, Review Rating, Shipping Type)

Tools Used: Python, PostgreSQL, SQL, Power BI

Missing Data: 54 values in Review Rating column

3. Exploratory Data Analysis using Python

We began with data preparation and cleaning in Python:

- **Data Loading:** Imported the dataset using pandas.
- **Initial Exploration:** Used `df.info()` to check structure and `.describe()` for summary statistics.

	Customer ID	Age	Gender	Item Purchased	Category	Unit sold	Unit Price	Purchase_Amount_inr	Location	Size	Color	Season	Review Rating	Subscrip St
count	4500.000000	4500.000000	4500	4500	4500	4500.000000	4500.000000	4500.000000	4500	4500	4500	4500	4446.000000	4
unique	NaN	NaN	2	25	4	NaN	NaN	NaN	60	4	25	4	NaN	
top	NaN	NaN	Male	Blouse	Clothing	NaN	NaN	NaN	Navi Mumbai	M	Olive	Spring	NaN	
freq	NaN	NaN	2920	207	2024	NaN	NaN	NaN	79	2032	214	1140	NaN	2
mean	2250.500000	44.340000	NaN	NaN	NaN	2.039556	2991.500000	6080.611111	NaN	NaN	NaN	NaN	3.754116	I
std	1299.182435	15.410958	NaN	NaN	NaN	0.886910	1182.413772	3673.234677	NaN	NaN	NaN	NaN	0.716666	I
min	1.000000	18.000000	NaN	NaN	NaN	1.000000	1000.000000	1000.000000	NaN	NaN	NaN	NaN	2.500000	I
25%	1125.750000	31.000000	NaN	NaN	NaN	1.000000	1950.000000	3300.000000	NaN	NaN	NaN	NaN	3.100000	I
50%	2250.500000	44.000000	NaN	NaN	NaN	2.000000	3000.000000	4950.000000	NaN	NaN	NaN	NaN	3.800000	I
75%	3375.250000	57.000000	NaN	NaN	NaN	3.000000	4012.500000	8500.000000	NaN	NaN	NaN	NaN	4.400000	I
max	4500.000000	75.000000	NaN	NaN	NaN	9.000000	5000.000000	27900.000000	NaN	NaN	NaN	NaN	5.000000	I

Subscription Status	Shipping Type	Discount Applied	Promo Code Used	Previous Purchases	Payment Method	Frequency of Purchases
4500	4500	4500	4500	4500.000000	4500	4500
2	6	2	2	NaN	12	7
No	Free Shipping	Yes	Yes	NaN	Credit Card	Annually
2847	782	2277	2277	NaN	639	679
NaN	NaN	NaN	NaN	25.628889	NaN	NaN
NaN	NaN	NaN	NaN	14.540791	NaN	NaN
NaN	NaN	NaN	NaN	1.000000	NaN	NaN
NaN	NaN	NaN	NaN	13.000000	NaN	NaN
NaN	NaN	NaN	NaN	26.000000	NaN	NaN
NaN	NaN	NaN	NaN	38.000000	NaN	NaN
NaN	NaN	NaN	NaN	50.000000	NaN	NaN

Missing Data Handling: Checked for null values and imputed missing values in the Review Rating column using the median rating of each product category.

- **Column Standardization:** Renamed columns to **snake case** for better readability and documentation.
- **Feature Engineering:**
 - Created **age_group** column by binning customer ages.
 - Created **purchase_frequency_days** column from purchase data.
- **Data Consistency Check:** Verified if discount_applied and promo_code_used were redundant; dropped promo_code_used.
- **Database Integration:** Connected Python script to PostgreSQL and loaded the cleaned DataFrame into the database for SQL analysis.

4. Data Analysis using SQL (Business Transactions)

We performed structured analysis in PostgreSQL to answer key business questions:

1. **Revenue by Category** – Compared total revenue generated across different product categories.

	category text	total_revenue text	total_orders bigint	avg_rating numeric
1	Clothing	12.34 M	2024	3.74
2	Accessories	8.61 M	1421	3.76
3	Footwear	4.17 M	671	3.80
4	Outerwear	2.24 M	384	3.75

2. Revenue by Location – Identified locations generating the highest total revenue.

	location text	total_revenue text
1	Guwahati	0.56 M
2	Kalyan-Dombivli	0.54 M
3	Rajkot	0.53 M
4	Faridabad	0.53 M
5	Jodhpur	0.52 M
6	Siliguri	0.52 M
7	Mangaluru	0.52 M
8	Gurugram	0.51 M
9	Kochi	0.51 M
10	Raipur	0.51 M
11	Vasai-Virar	0.51 M
12	Dehradun	0.50 M
13	Noida	0.50 M
14	Udaipur	0.50 M
15	Kolhapur	0.50 M
16	Navi Mumbai	0.49 M
17	Amritsar	0.48 M
Total rows: 60		Query complete 00:00:00

3. Top 5 Products by Average Rating – Identified the five products with the highest average customer review ratings.

	products text	Average Product Rating numeric
1	Gloves	3.84
2	Sandals	3.83
3	Boots	3.81
4	T-shirt	3.79
5	Socks	3.78

4. Top 5 Payment Methods by Revenue – Identified the five payment methods contributing the most to total sales.

	payment_method text	total_revenue text
1	Credit Card	3.87 M
2	Debit Card	3.77 M
3	Cash	3.12 M
4	Amazon Pay	2.52 M
5	Google Pay	2.29 M

5. **Revenue By Season** – Compared total revenue generated across different seasons to identify the highest-performing season.

	season text	total_revenue text
1	Fall	7.04 M
2	Spring	7.03 M
3	Winter	6.69 M
4	Summer	6.60 M

6. **Average Spend by Subscription Status** – Compared average purchase amounts of subscribed and non-subscribed customers.

	subscription_status text	Total customers bigint	Avg Spend text	Total Revenue text
1	Yes	1653	6.10K	10.09M
2	No	2847	6.07K	17.27M

7. **Revenue by Gender** – Compared total revenue generated by male vs. female customers.

	gender text	Revenue (INR) text
1	Female	9.56M
2	Male	17.80M

8. **Customer Segmentation** – Analyzed the distribution of customers across New, Returning, and Loyal segments.

	customer_segment text	Number of Customers bigint
1	Loyal	3597
2	Returning	813
3	New	90

9. **Purchase Frequency Distribution** – Analyzed the distribution of customers across different purchase frequency groups to understand recurring shopping patterns.

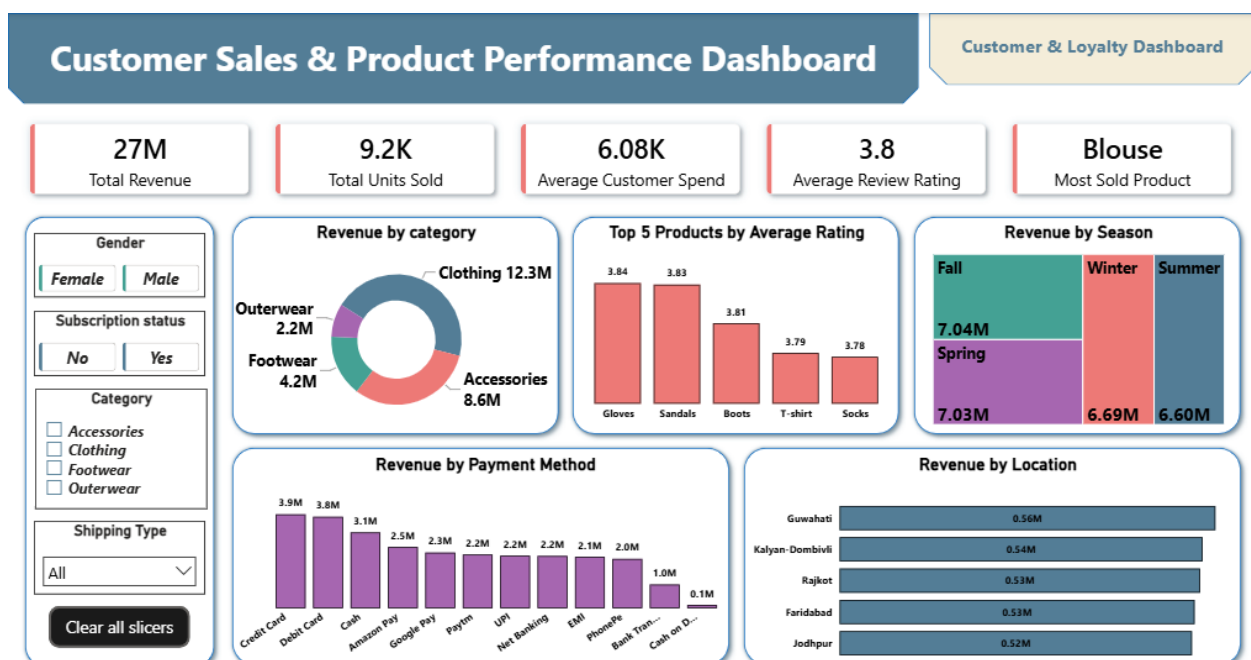
	frequency_of_purchases text	total_customers bigint	percentage text
1	Annually	679	15.09%
2	Every 3 Months	669	14.87%
3	Quarterly	634	14.09%
4	Fortnightly	633	14.07%
5	Weekly	631	14.02%
6	Monthly	631	14.02%
7	Bi-Weekly	623	13.84%

10. **Revenue by Age Group** – Compared total revenue generated across different customer age groups.

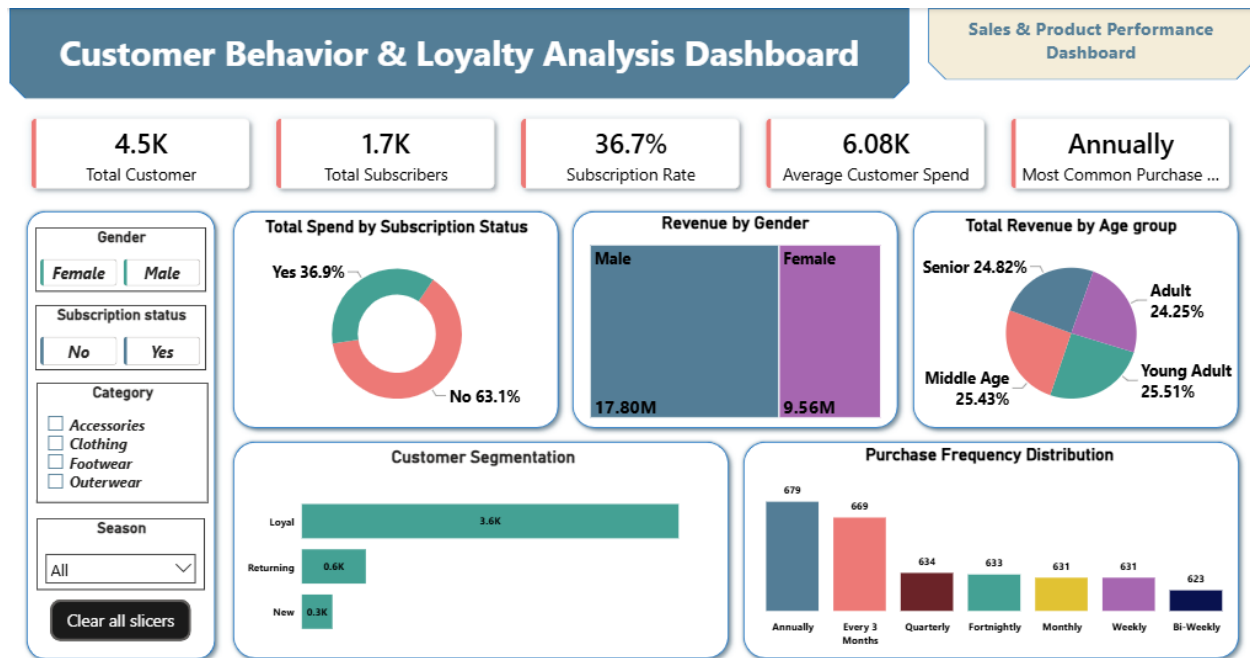
	age_group text	total_revenue text
1	Young Adult	6.98M
2	Middle Age	6.96M
3	Senior	6.79M
4	Adult	6.63M

5. Dashboards in Power BI

Dashboard 1 : Customer Sales & Product Performance



Dashboard 2 : Customer Behavior & Loyalty Analysis



6. Business Recommendations

- **Optimize Inventory** – Stock high-demand products to prevent missed sales.
- **Cross-Sell Products** – Bundle popular products with lower-performing categories.
- **Personalize Offers** – Use customer behavior and demographics for targeted promotions.
- **Improve Customer Experience** – Leverage ratings and feedback to improve products and services.
- **Expand Market Reach** – Target underperforming locations with focused marketing campaigns.
- **Optimize Payments** – Prioritize popular payment options for a smoother checkout experience.